Report R12 — the sectors in the matrix: block-diagonal (WCC) and Frobenius normal (SCC) form

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-08-04

Abstract

Every fragmentation statement in R2 and R9 is a number: a sector count, a growth base, a sub-leading exponent. Those numbers describe a property of a matrix — that the one-step transition matrix is block-diagonal in a basis nobody had drawn — and this report draws it. Two spy plots of the *same* matrix for rule 156 (`IIVI`) at `obc0`: once in the computational basis, once with the basis permuted so that each weak component is contiguous. The permuted picture is a staircase of 144 disjoint blocks at $N = 10$, the largest holding 20 of 1024 states, and **not one nonzero lies outside a block** — checked, at every N from 8 to 12, rather than asserted. Three quantitative readings come out of the same computation: the block count is exactly the Fibonacci number F_{N+2}, which is where R2's golden ratio comes from; the matrix thins from 1.81% to 0.24% dense between $N = 8$ and $N = 12$ while the *interior* of the blocks stays roughly two-thirds full, so the concentration factor grows from 41× to 259×; and the computational-basis panel is emphatically not structureless — it is strongly banded and self-similar — it simply does not show *sector* structure.

§4 then reorders the same matrix into its *Frobenius normal form* — strongly connected components in reverse topological order — for rules 156, 157, 109, 150 and 158, chosen to span both axes: exponential sector counts (φ, ρ, ψ) and polynomial ones, unitary and dissipative. Nothing lies below the block diagonal in any of them. For the two unitary rules the triangular form *is* the diagonal form (R9's A2, drawn); for the three dissipative ones the strictly upper part is most of the matrix — 95% of it for rule 158. The pair 150/158 is the sharpest case: they share the *identical sector partition* at every N tested, so no sector observable can distinguish them, while their Frobenius forms are 6 strong components against 936. `obc0` only; `pbc` is held.

Contents

1 What is plotted, and why

Definition 1 (the matrix)**.** For a rule, a length N and a boundary condition, let

$$M_{yx} = 1 \iff \text{the one-step map carries } |x\rangle \text{ to } |y\rangle \text{ with nonzero amplitude.}$$

M is the support of the circuit unitary U, equivalently of the doubly stochastic $|U|^2$; the two have the same pattern, which is all a spy plot uses. Amplitudes are deliberately discarded — the point of the figure is that the *support alone* already carries the fragmentation.

Definition 2 (the permutation)**.** Let the weak components of the transition graph be $S_1, \dots, S_K$, ordered by size, largest first, with states ascending inside each. Let P be the permutation that lists the basis in that order. The second panel plots PMP^{-1}.

Why rule 156**.** It is unitary (`IIVI`: two identities, a Hadamard, an identity), so $|U|^2$ is doubly stochastic, there are no transient states, and the weak, strong and forward-closure partitions all coincide — R9's assertion A2. The blocks are therefore unambiguous, and there is no question of whether we are drawing sectors or attractors: for this rule they are the same objects. It is also the rule R2 solves by room-packing, so its numbers are known independently of anything here.

Why it is worth a figure. "The dynamics block-diagonalises" is the central claim of the whole sector programme, and until now it has appeared only as counts and exponents. A permutation is not a theorem, but it makes the claim visible in one glance, and it makes the *failure* of the computational basis to reveal it equally visible.

2 The two panels

Three things to read off Figure 1.

- **The sorted panel is exactly block-diagonal.** Every one of the 6930 nonzeros lies inside one of the 144 shaded squares. Nothing is off-diagonal, at any N tested (§3).
- **The blocks are small and there are many of them.** The largest holds 20 of 1024 states, so the staircase is a thin diagonal thread rather than a few fat blocks. That is $D_{\max}/2^N \approx 0.02$ and falling — the quantitative form of "strong fragmentation", and the reason R2's $b = 4^{1/5}$ sits so far below the ergodic $b = 2$.
- **The left panel is *not* structureless.** It is strongly banded, with a self-similar block-of-blocks pattern at several scales. That is real and has an explanation — the integer order groups states by their high bits, and the rule is local, so states that agree on most sites sit near one another — but the bands run *across* sectors. Ordering by integer value reveals locality, not conservation. The contrast between the panels is not structure-versus-noise; it is the wrong structure versus the right one.

2.1 A closer look

At $N = 10$ the largest block is 20 states wide, which is about four pixels in Figure 1, so the interior of a block cannot be seen there. Figure 2 takes the first 140 rows and columns.

The blocks are *not* full: within a block the fill is about two-thirds (§3), and the interior pattern is itself structured rather than random. So a sector is not a featureless invariant subspace — it has its own sparse combinatorics, which is what the room-packing derivation of R2 is about. A smaller system makes the same point without a zoom:

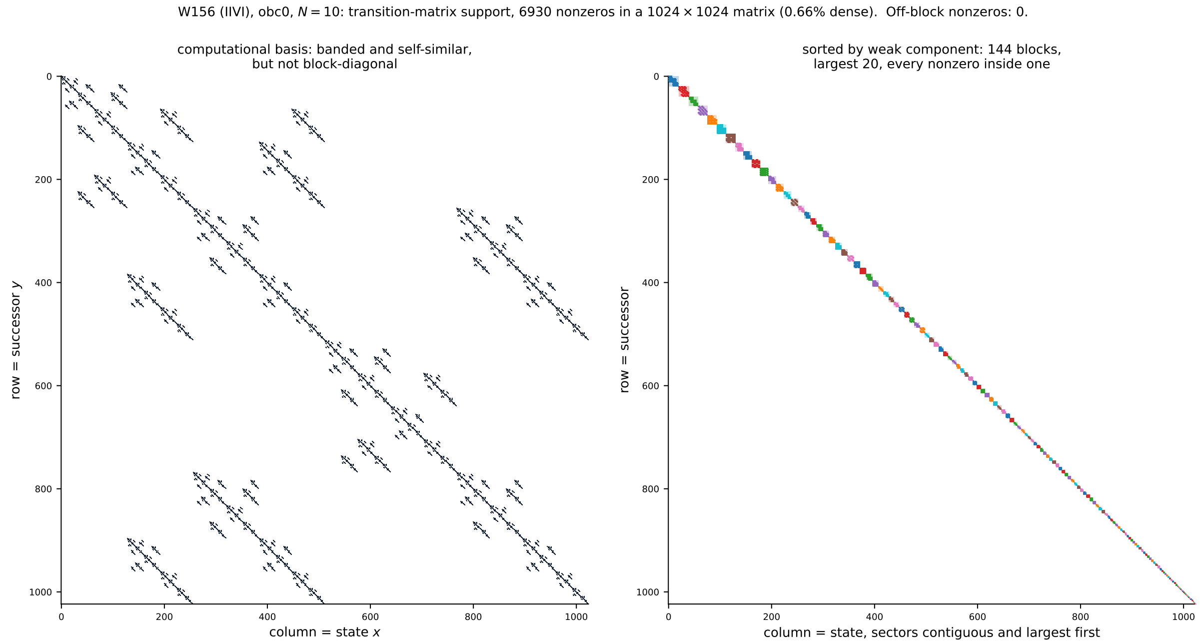

Figure 1: **S1, the same matrix in two bases** (rule 156, `obc0`, $N = 10$; 1024×1024, 6930 nonzeros, 0.66% dense). **Left:** computational basis, i.e. states ordered by their integer value. **Right:** basis permuted so each weak component is contiguous, largest first; the shaded squares are the components and the dots are coloured by which component they belong to. The two panels contain *identical* data. The right panel's caption number — off-block nonzeros = 0 — is computed, not assumed.

3 Numbers

N	dim	nnz	density	blocks	$D_{\max}$	off-block
8	256	1189	1.81%	55	12	0
9	512	2870	1.09%	89	16	0
10	1024	6930	0.66%	144	20	0
11	2048	16731	0.40%	233	27	0
12	4096	40391	0.24%	377	36	0

Off-block nonzeros are zero at every size. That column is the whole content of the sorted panel, and it is a computation (`spy.check_block_diagonal`), not a restatement of the definition: the blocks are found by weak connectivity on the transition graph, and the check then asks independently whether any matrix entry crosses between them. A nonzero there would mean the partition and the matrix disagree.

The block count is Fibonacci. For $N = 6 \dots 12$ the number of sectors is

$$21,\ 34,\ 55,\ 89,\ 144,\ 233,\ 377\ =\ F_{N+2},$$

exactly, with $F_1 = F_2 = 1$. Hence the growth base $\varphi = 1.618034$ that R2 reports for this rule's sector count at `obc0`. (Under `pbc` the same rule gives $L_N + 1$ with L the Lucas numbers — a different sequence with the same base, which is why the boundary condition has to be stated with the constant.)

The nonzero count is a third constant, exactly. Counting the edges rather than the sectors gives, for $N = 6 \dots 14$,

$$204,\ 493,\ 1189,\ 2870,\ 6930,\ 16731,\ 40391,\ 97512,\ 235416,$$

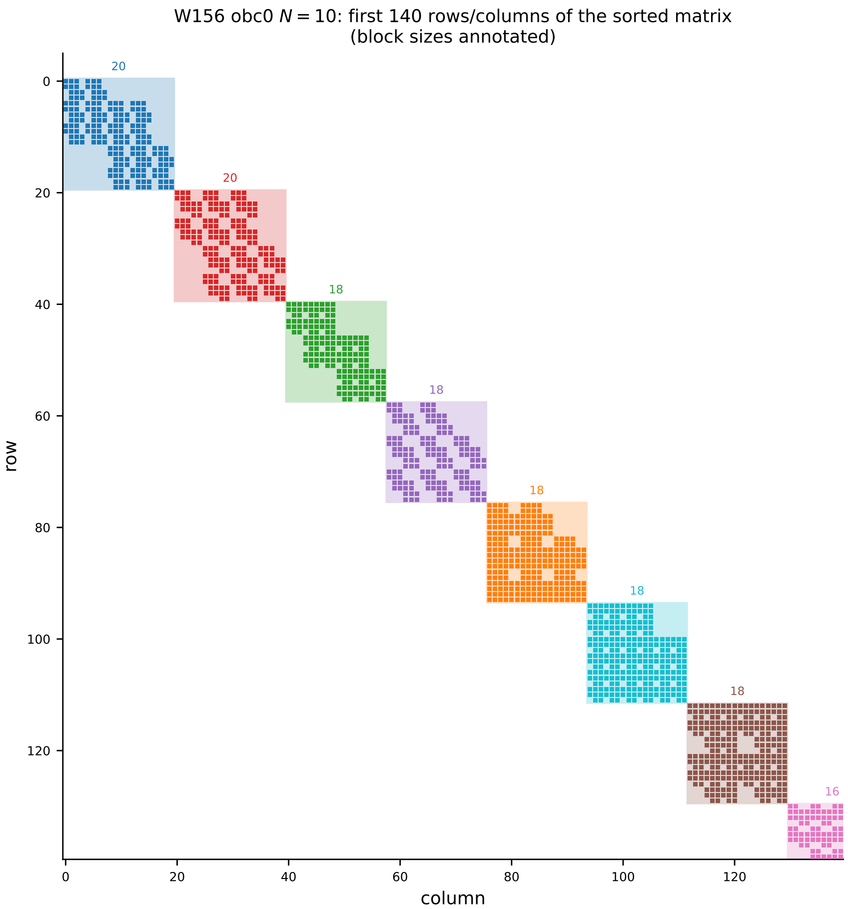

Figure 2: **S2, the top-left corner of the sorted matrix** (rule 156, `obc0`, $N = 10$), block sizes annotated. The largest sectors are 20, 20, 18, 18, 18, 18, 18, 16, . . . states. Each block has its own internal sparsity pattern, itself visibly self-similar.

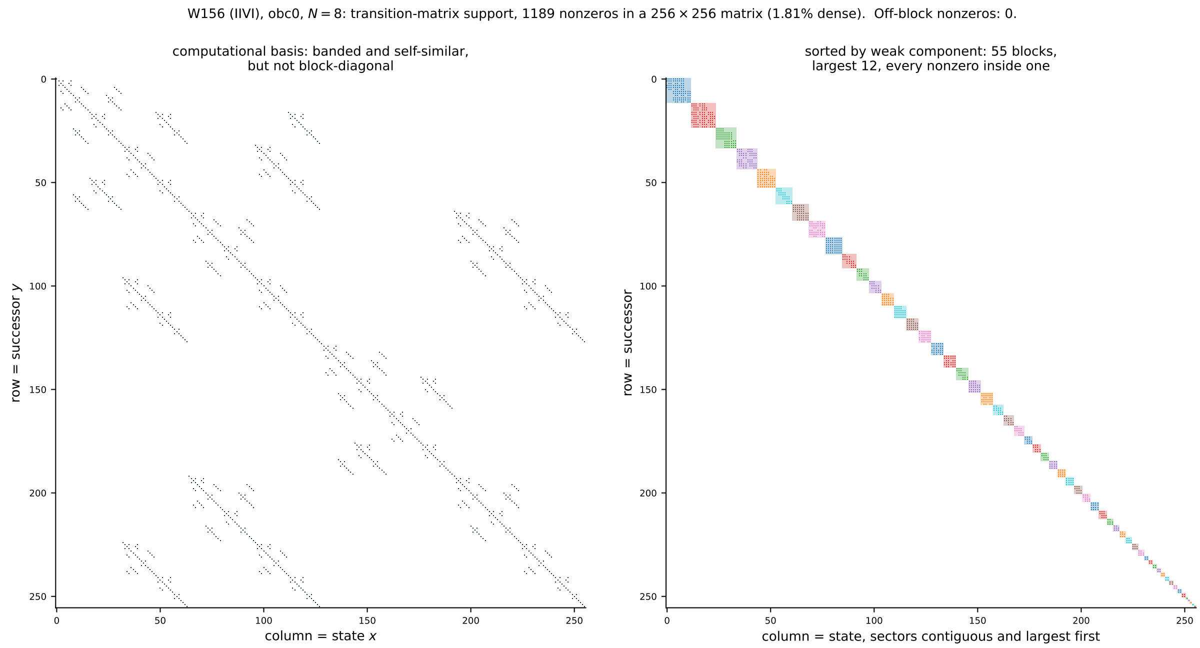

Figure 3: **S3, the same pair at** $N = 8$ (256×256, 55 blocks, largest 12). Small enough that the block structure is legible in the full matrix.

which satisfies the integer recurrence

$$a_N \;=\; 2a_{N-1} + 2a_{N-3} + a_{N-4},$$

whose characteristic polynomial factors as

$$x^4 - 2x^3 - 2x - 1 \;=\; (x^2 - 2x - 1)(x^2 + 1).$$

The dominant root is the *silver ratio* $1 + \sqrt{2} = 2.414214$, and the $\pm i$ from the second factor is a pure period-4 modulation — which is exactly the $-1, -1, +1, +1$ pattern by which a_N misses $2a_{N-1} + a_{N-2}$. So this one rule carries three different algebraic constants at once, one for each question asked of the same matrix: φ for how many blocks there are, $4^{1/5}$ for how big the largest gets (R2), and $1 + \sqrt{2}$ for how many nonzeros the matrix has. None of the three is a fit; all are exact recurrences or derivations.

Thinning globally, not locally. As N grows the matrix becomes rapidly sparser in absolute terms, but the blocks do not hollow out:

N	density of M	fill *inside* the blocks	ratio
8	1.81%	74.1%	41×
10	0.66%	68.0%	103×
12	0.24%	62.3%	259×

The middle column is the nonzero count divided by $\sum_k |S_k|^2$, the total area available inside the diagonal blocks. It drifts down slowly while the global density falls by a factor of 7.5, so essentially all of the sparsity of M is the block structure and almost none of it is sparsity within a sector. That is the precise sense in which the permutation is the right change of basis: it accounts for the emptiness.

4 The Frobenius normal form

The WCC permutation makes M block-*diagonal*. The finer SCC partition makes it block-*triangular*, which is the Frobenius (irreducible) normal form of a non-negative matrix. The two coincide exactly when there are no transient states, and that is the same condition as unitarity — so this section is where the unitary/dissipative distinction becomes a picture.

Definition 3 (Frobenius order)**.** Take the strongly connected components of the transition graph and order them by a *reverse* topological order of the condensation: sinks — the terminal SCCs, i.e. the monitored attractors — first, sources last, larger first among those free to move. An edge then runs from a higher index to a lower one, so every nonzero sits on or *above* the block diagonal. The attractors occupy the top-left corner and everything drains up and to the left.

The ordering is produced by a Kahn sweep on the condensation (`spy.scc_blocks`) rather than by trusting Tarjan's internal numbering. Tarjan does happen to finalise a component before its predecessors, but that is an implementation detail of the traversal, and the triangularity of the figure should not depend on it; `spy.check_frobenius` then verifies the result independently by counting the nonzeros below the block diagonal.

rule	tuple	n_{wcc}	n_{scc}	terminal	$\lvert\text{Rec}\rvert$	in blocks	upper	below
156	`IIVI`	144	144	144	1024	6930	0	0
157	`EIVI`	16	758	16	67	1733	5414	0
109	`EIIV`	54	723	54	355	5131	6042	0
150	`IVVI`	6	6	6	1024	59049	0	0
158	`IEVI`	6	936	6	31	374	7545	0

(`obc0`, $N = 10$. "in blocks" counts nonzeros inside an SCC, "upper" the strictly upper part — the drainage — and "below" is the number that would falsify the triangular form.)
The five rules were chosen to span the two axes that matter. Sector count: exponential for 156 (φ), 157 (ρ) and 109 (ψ); polynomial (linear) for 150 and 158. Dynamics: unitary for 156 and 150, dissipative for 157, 109 and 158.

4.1 What the three panels show

- **Nothing lies below the block diagonal, for any of the five.** The Frobenius form is genuinely triangular, computed and not assumed.
- **For a unitary rule the triangular form *is* the diagonal form.** Rules 156 and 150 have $n_{\text{scc}} = n_{\text{wcc}}$, every SCC terminal, all 1024 states recurrent and a drainage of exactly zero. Their third panel is their second panel with the shading removed. That is R9's assertion A2 as a picture: with $|U|^2$ doubly stochastic there is nowhere for weight to drain to, so weak and strong components cannot differ.
- **For a dissipative rule they are nothing alike.** Rule 158 has 6 weak components and 936 strong ones; 157 has 16 and 758; 109 has 54 and 723. The recurrent set is a thin gold corner — 31 states of 1024 for 158, 67 for 157 — and the bulk of the matrix is the grey drainage cloud filling the upper triangle. For 158, 7545 of 7919 nonzeros (95%) are strictly upper: almost the entire matrix is transient flow, and only 5% of it is the dynamics *inside* an attractor.
- **One attractor per sector, in all five.** $n_{\text{terminal}} = n_{\text{wcc}}$ exactly for every rule here, which is the generic case of R9 §7.3 — none of these five is among the twelve rules whose sectors carry more than one attractor. So the coarsest partition (WCC) and the set of sinks are in bijection, and everything between them is transient scaffolding.

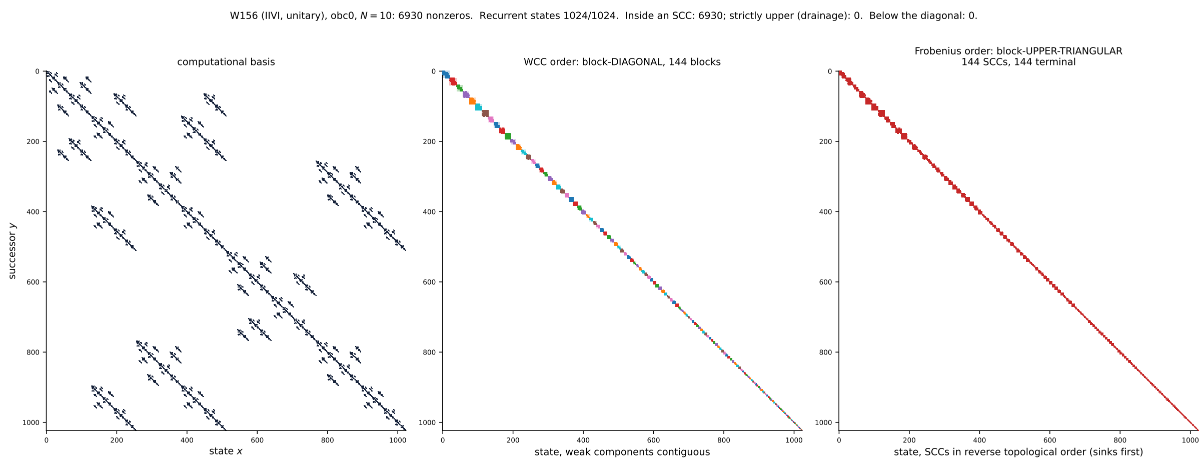

Figure 4: **S4, rule** 156 **(`IIVI`, unitary, φ).** Left: computational basis. Middle: WCC order, block-diagonal. Right: Frobenius order. All 144 SCCs are terminal, so the third panel is block-diagonal too and the drainage is empty — for a unitary rule the two normal forms coincide.

4.2 Rules 150 and 158: the same sectors, different matrices

This pair is the sharpest statement in the report. Rules 150 and 158 do not merely have the same sector *count*: at $N = 8$, 10 and 12 they have the *identical sector partition*, the same sets of states, with sizes $462, 330, 165, 55, 11, 1$ at $N = 10$. Their middle panels are the same picture.

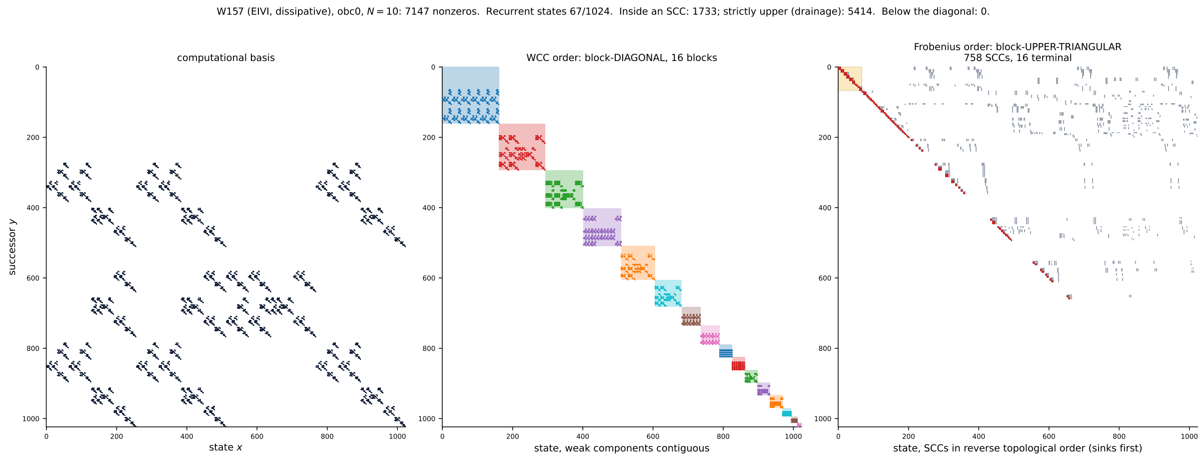

Figure 5: **S5, rule** 157 (EIVI**, dissipative,** ρ**).** 16 weak components but 758 strong ones. Gold marks the recurrent corner (67 states); red are the entries inside an SCC, grey the drainage.

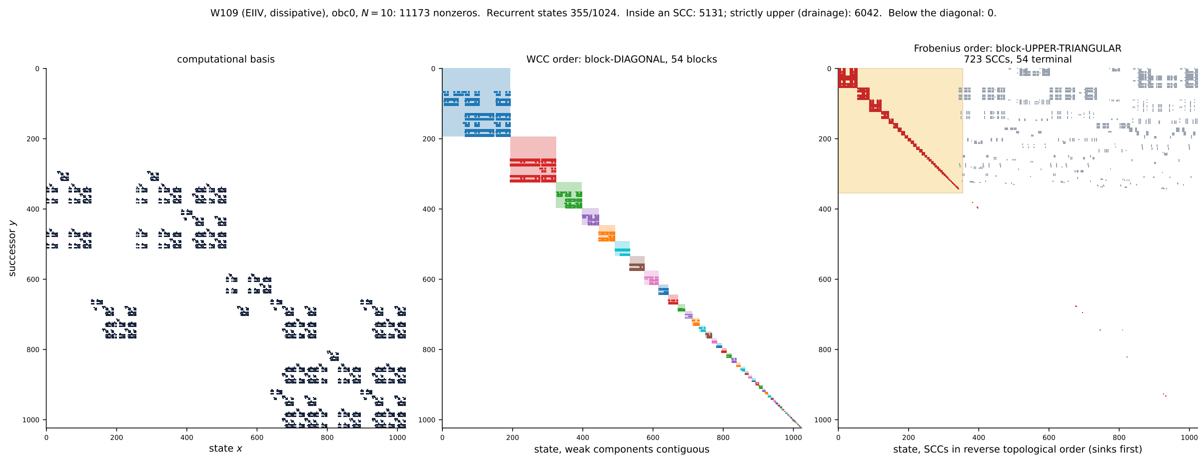

Figure 6: **S6, rule** 109 (EIIV**, dissipative,** ψ**).** The least contracting of the three dissipative rules: 355 of 1024 states recurrent and a largest SCC of 55, so the gold corner is a third of the matrix and the diagonal blocks are visible in their own right.

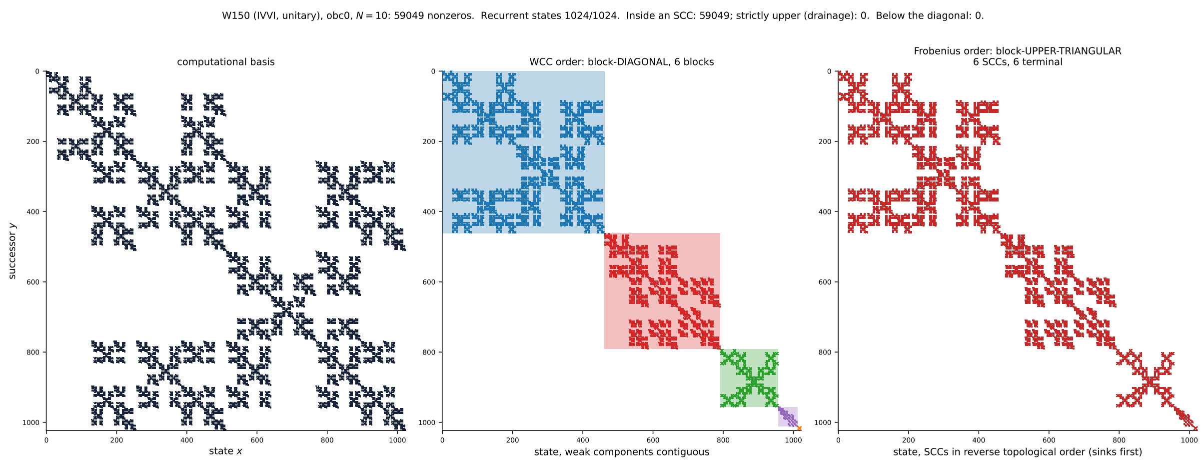

Figure 7: **S7, rule** 150 (IVVI**, unitary, linear sector count).** Six sectors of sizes $462, 330, 165, 55, 11, 1$; every one is a single SCC.

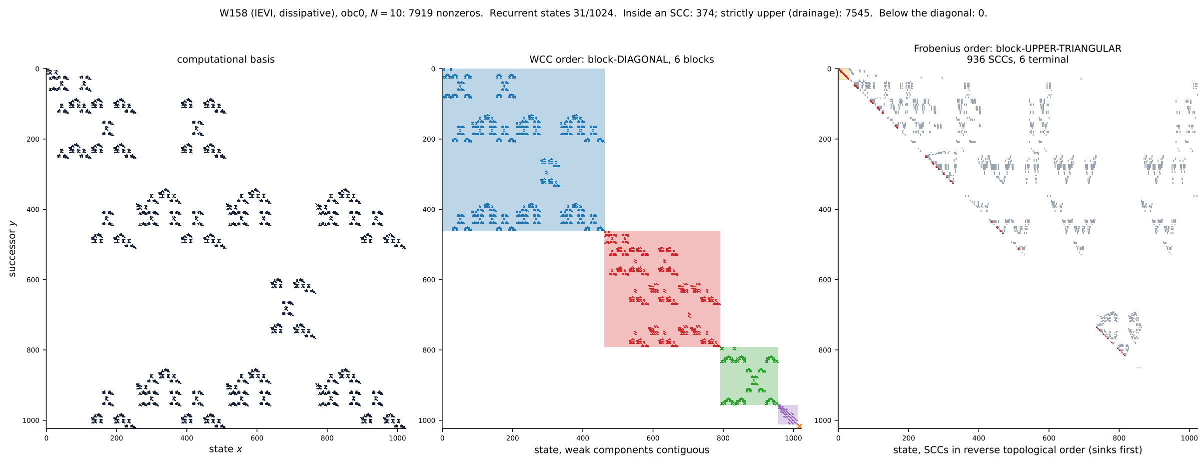

Figure 8: **S8, rule** 158 **(**`IEVI`**, dissipative, linear sector count).** The *same six sectors as rule* 150 — middle panel — shattered into 936 strong components of which 6 are terminal, holding 31 states between them.

Yet one is unitary with all 1024 states recurrent in 6 strong components, and the other retains 31 recurrent states in 6 terminal components out of 936. Nothing on the sector axis — not the count, not the sizes, not the partition itself — can tell them apart; the Frobenius form separates them completely. That is the concrete form of R9's finding that the sector axis is uninformative for the V+reset family, and the reason this project carries two maps rather than one.

5 What this does and does not show

- **It is support, not dynamics.** A spy plot cannot distinguish a sector that the dynamics explores ergodically from one it merely fails to leave. The block structure is a statement about which amplitudes can be nonzero, and nothing here measures how the weight moves inside a block.
- **It is one rule and one boundary condition.** Rule 156 was chosen because unitarity makes the partition unambiguous (A2). For a dissipative rule the same picture would need a choice — weak components block-*diagonal*, strong components only block-*triangular* — which is the distinction adjudicated in R-T14 and quantified in R9 §7.3. `pbc` is held pending caveats and is not drawn.
- **The permutation is not canonical.** Blocks are ordered by size and states ascending within a block; any other order inside a sector gives an equally valid block-diagonal picture with a different interior pattern. What is canonical is the *partition*, and therefore the block sizes.
- **It does not prove the asymptotics.** The figure is a finite-N object. That the block count is F_{N+2} is checked to $N = 12$ here and derived in R2; the picture illustrates it and does not establish it.

6 Reproduce

```
python -m qca_fragmentation.viz.spy --rule 156 --N 10 --bc obc0
```

writes `fig_spy_156_N10_obc0`, `fig_spy_156_N8_obc0`, `fig_spy_zoom_156_N10_obc0` (each as `.pdf` and `.png`) and `reports/tex/tab_r12_spy_obc0.tex`. The module is `code/qca_fragmentation/viz/spy.py`;

its tests are in `code/qca_fragmentation/tests/test_spy.py`, and they pin the zero off-block count, the Fibonacci block count, the partition agreeing with the Tier-1e `weak_components` pass, and the permutation being a genuine permutation.